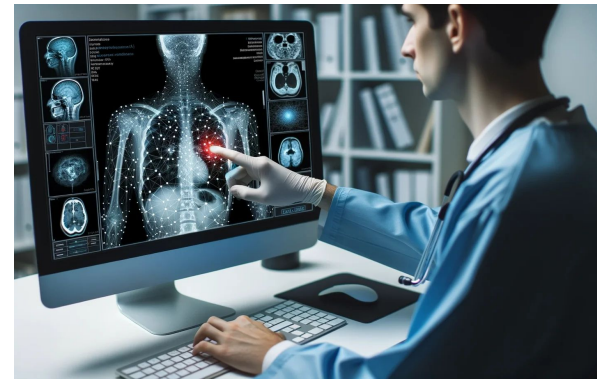
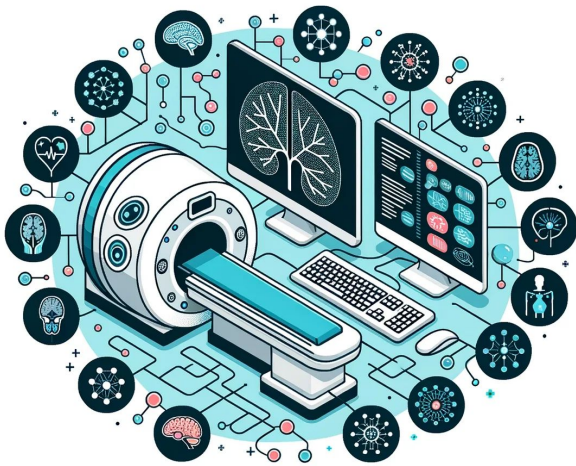


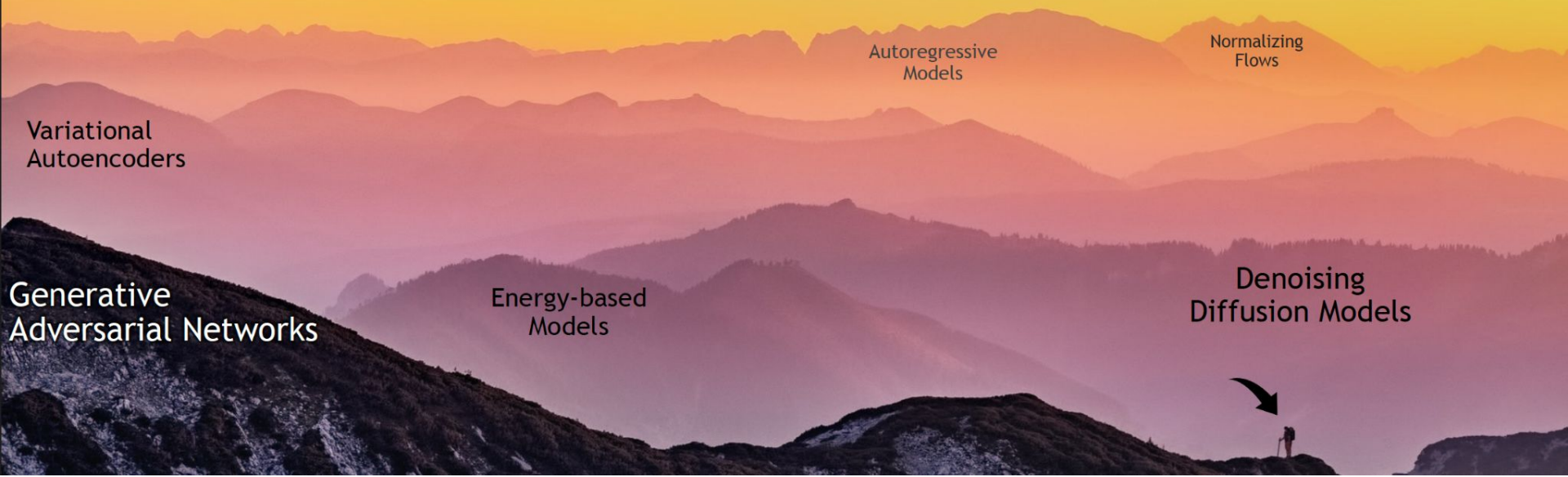
Applied Deep Learning and Generative Models in Healthcare



Session 5: Medical Image Synthesis with Diffusion
Date: Mar 15 2025

Instructor: Mahmoud E. Khani, Ph.D.

The Landscape of Deep Generative Learning



Variational
Autoencoders

Autoregressive
Models

Normalizing
Flows

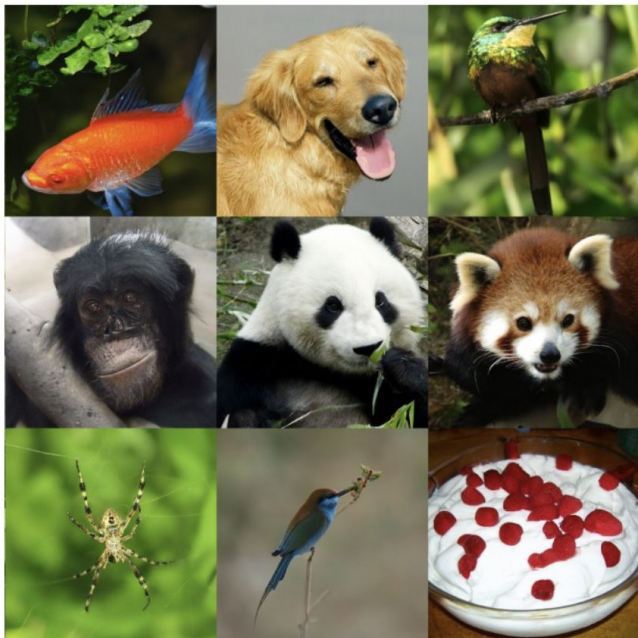
Generative
Adversarial Networks

Energy-based
Models

Denoising
Diffusion Models



Denoising Diffusion Models



[“Diffusion Models Beat GANs on Image Synthesis”](#)
Dhariwal & Nichol, OpenAI, 2021



[“Cascaded Diffusion Models for High Fidelity Image Generation”](#)
Ho et al., Google, 2021

Text-to-Image Generation

DALL·E 2

“a teddy bear on a skateboard in times square”



[“Hierarchical Text-Conditional Image Generation with CLIP Latents”](#)
Ramesh et al., 2022

Imagen

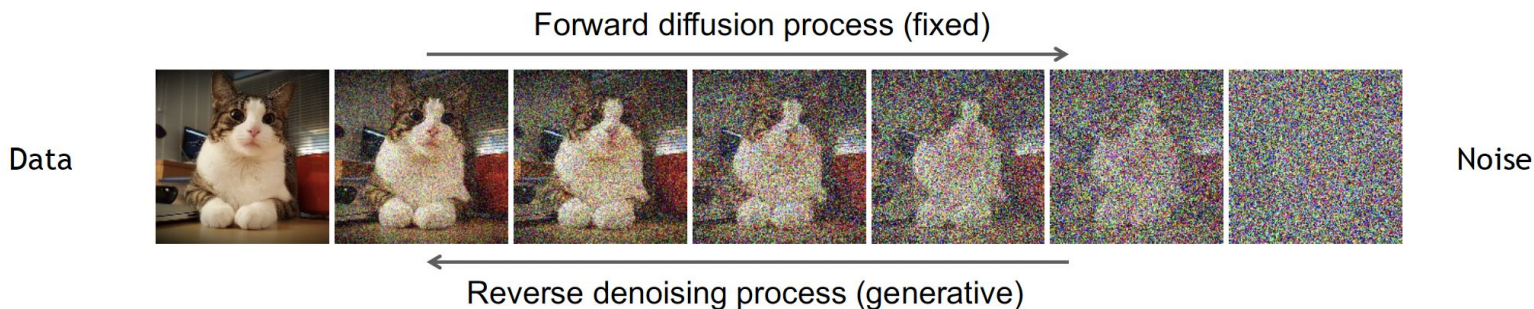
A group of teddy bears in suit in a corporate office celebrating the birthday of their friend. There is a pizza cake on the desk.



[“Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding”](#), Saharia et al., 2022

Learning to Generate by Denoising

- Denoising diffusion consists of two processes:
 - Forward diffusion to gradually add noise to input
 - Reverse denoising that learns to generate data by denoising



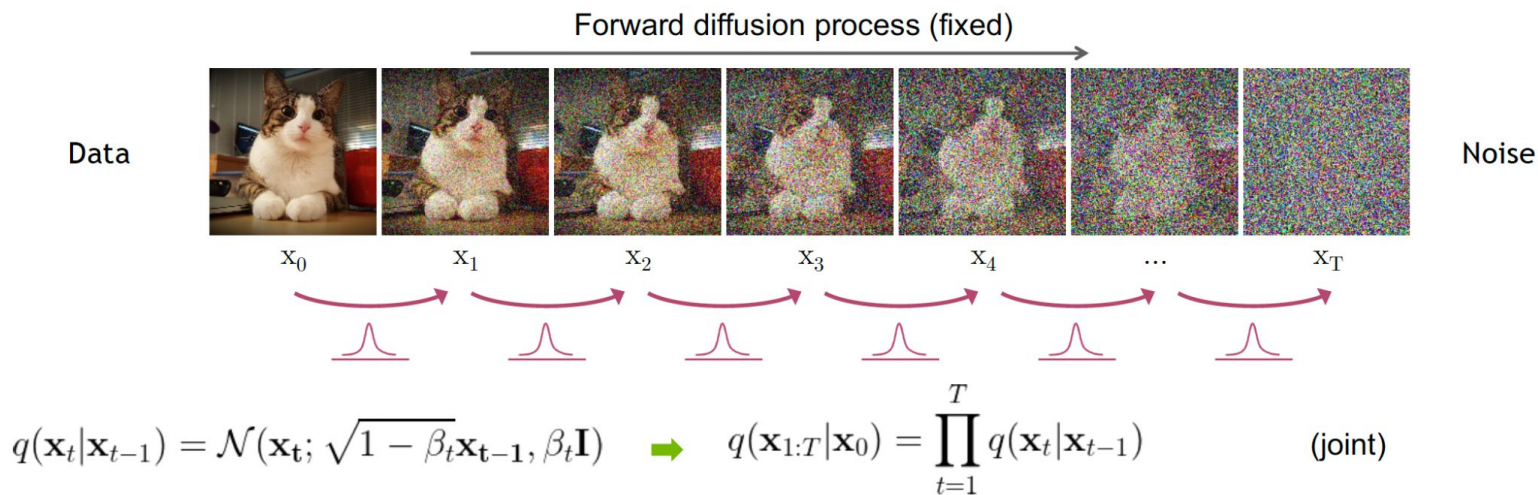
[Sohl-Dickstein et al., Deep Unsupervised Learning using Nonequilibrium Thermodynamics, ICML 2015](#)

[Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020](#)

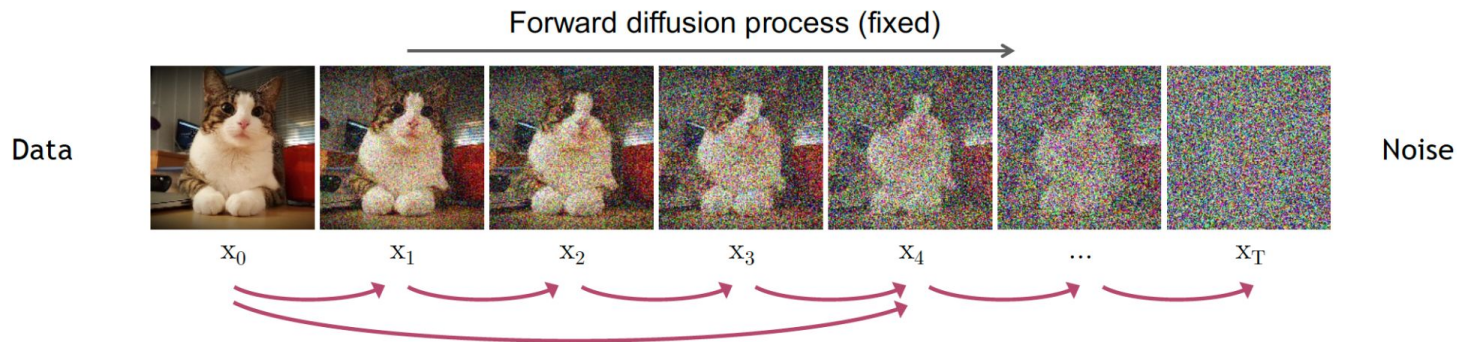
[Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021](#)

Forward Diffusion Process

- The formal definition of the forward process in T steps:



Diffusion Kernel



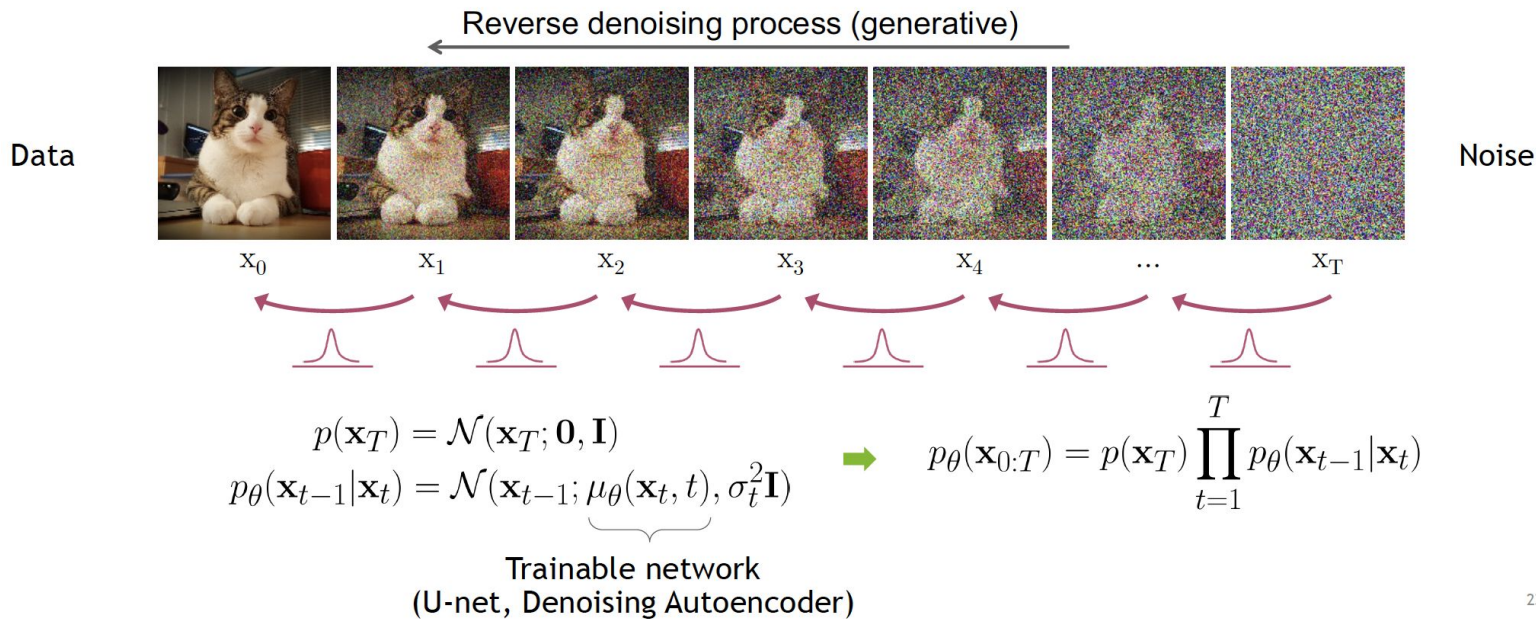
Define $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ $\rightarrow q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$ (Diffusion Kernel)

For sampling: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{(1 - \bar{\alpha}_t)} \epsilon$ where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

β_t values schedule (i.e., the noise schedule) is designed such that $\bar{\alpha}_T \rightarrow 0$ and $q(\mathbf{x}_T | \mathbf{x}_0) \approx \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$

Reverse Denoising Process

- Formal definition of reverse process in T steps:



Training and Sampling Algorithms

Algorithm 1 Training

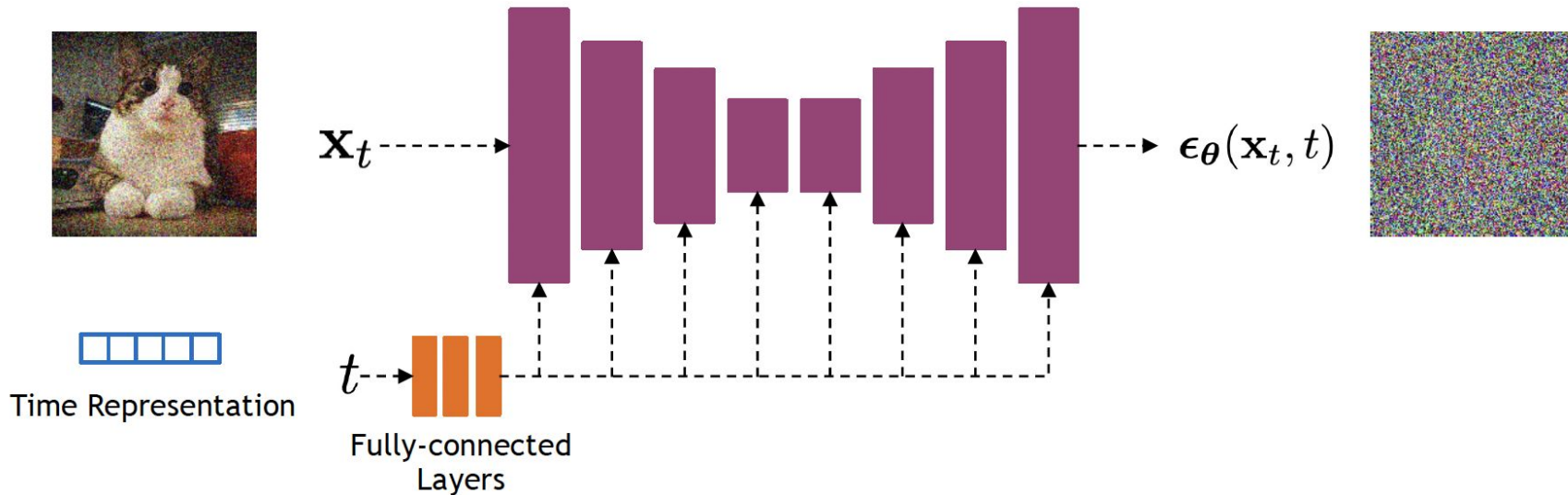
```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$   
6: until converged
```

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

Network Architectures

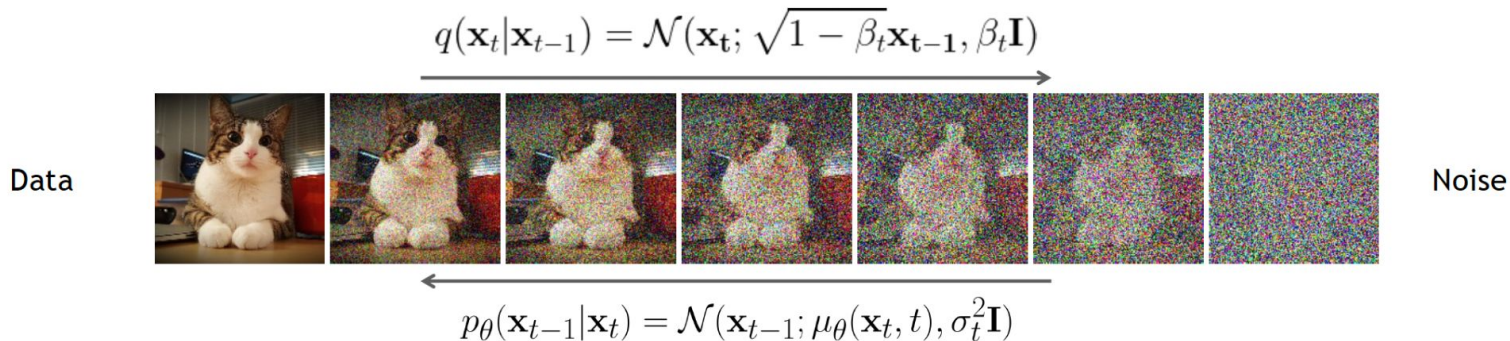
- Diffusion models often use **U-Net** architectures with **ResNet** blocks and **self-attention** layers



Time representation: sinusoidal positional embeddings or random Fourier features.

Diffusion Parameters

- The noise schedule



Above, β_t and σ_t^2 control the variance of the forward diffusion and reverse denoising processes respectively.

Often a linear schedule is used for β_t , and σ_t^2 is set equal to β_t .

Fourier Analysis of the Forward Process

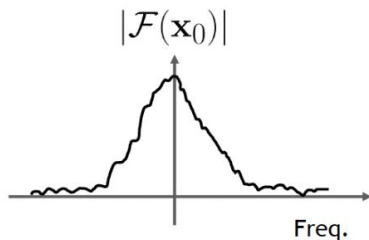
Recall that sampling from $q(\mathbf{x}_t|\mathbf{x}_0)$ is done using $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{(1 - \bar{\alpha}_t)} \epsilon$ where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{(1 - \bar{\alpha}_t)} \epsilon$$

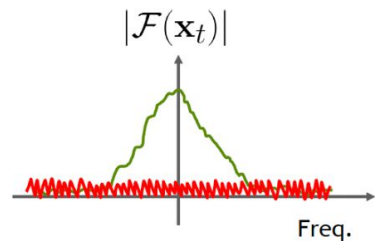


Fourier Transform

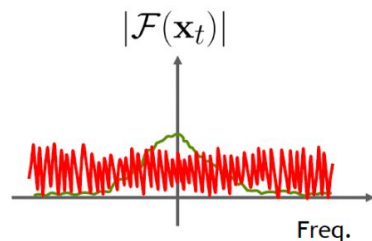
$$\mathcal{F}(\mathbf{x}_t) = \sqrt{\bar{\alpha}_t} \mathcal{F}(\mathbf{x}_0) + \sqrt{(1 - \bar{\alpha}_t)} \mathcal{F}(\epsilon)$$



Small t
 $\bar{\alpha}_t \sim 1$



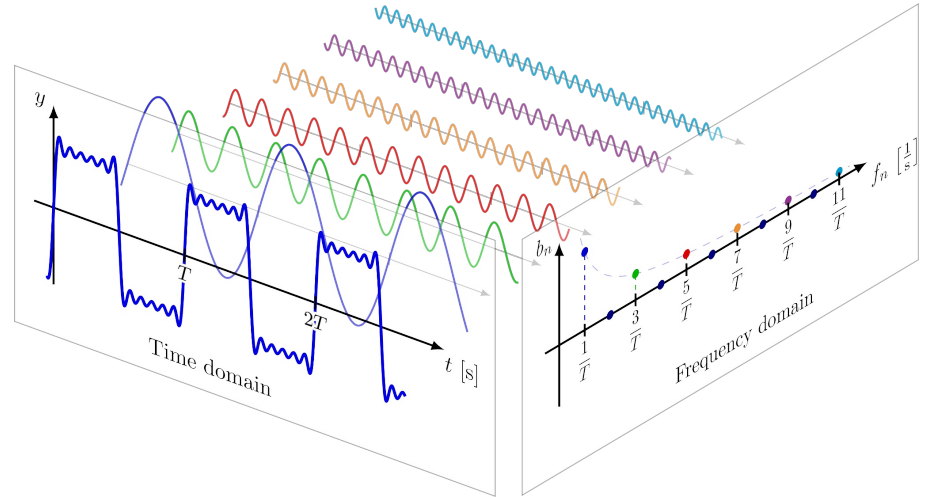
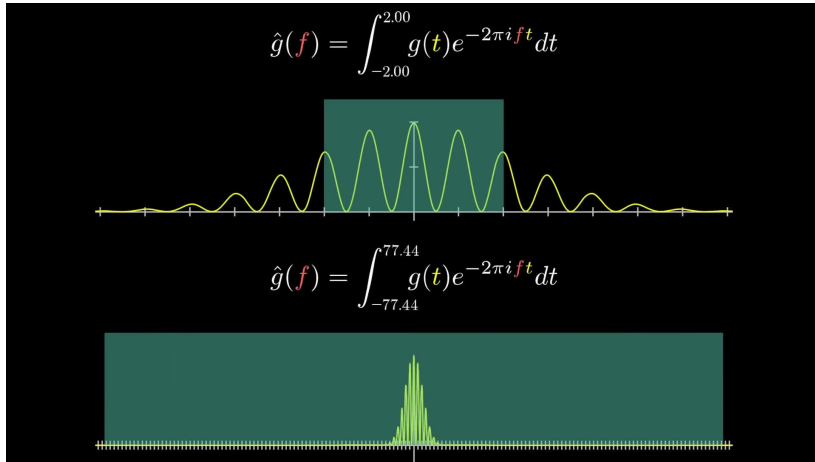
Large t
 $\bar{\alpha}_t \sim 0$



In the forward diffusion, the high frequency content is perturbed faster.

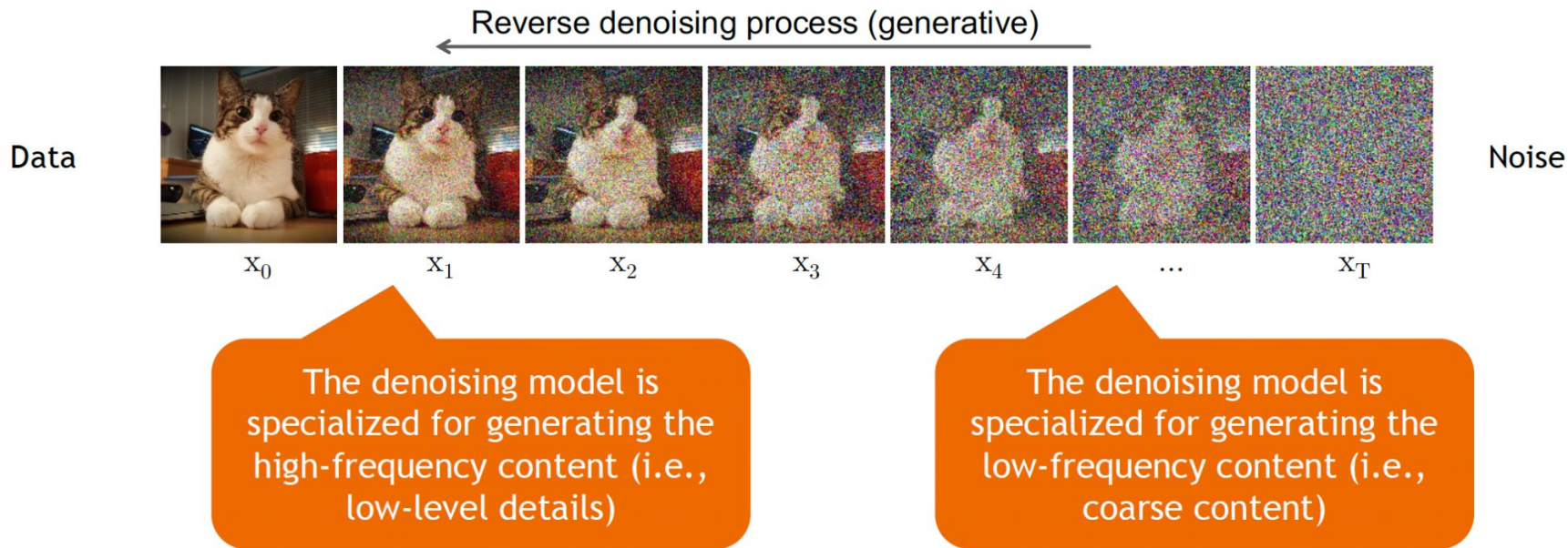
Fourier Analysis

- Decomposition of signal into **frequency** components



<https://dibsmethodsmeetings.github.io/fourier-transforms/>

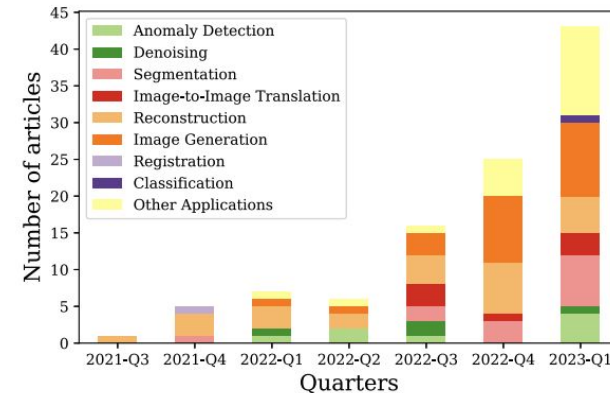
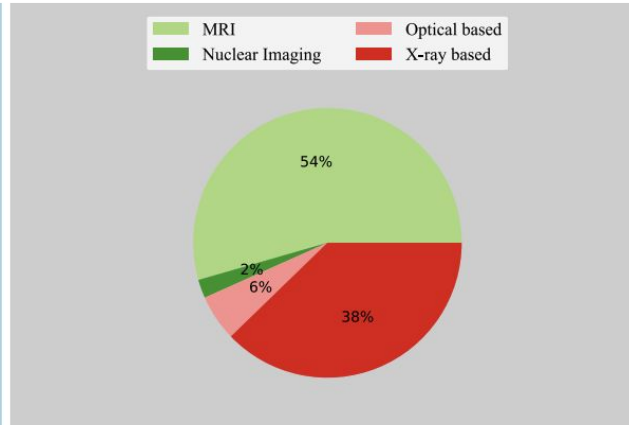
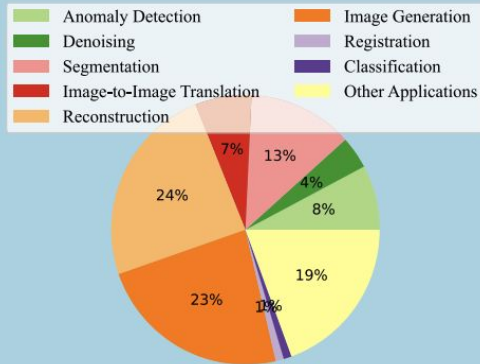
Content-Detail Tradeoff



The weighting of the training objective for different timesteps is important!

Diffusion Models in Medical Imaging

- Diffusion models have emerged as **powerful generative models** in medical imaging
 - improved sample quality, mode coverage, and versatility across various applications.



**MedAI
Group**



Session #96: Denoising Diffusion Models for Medical Image Analysis

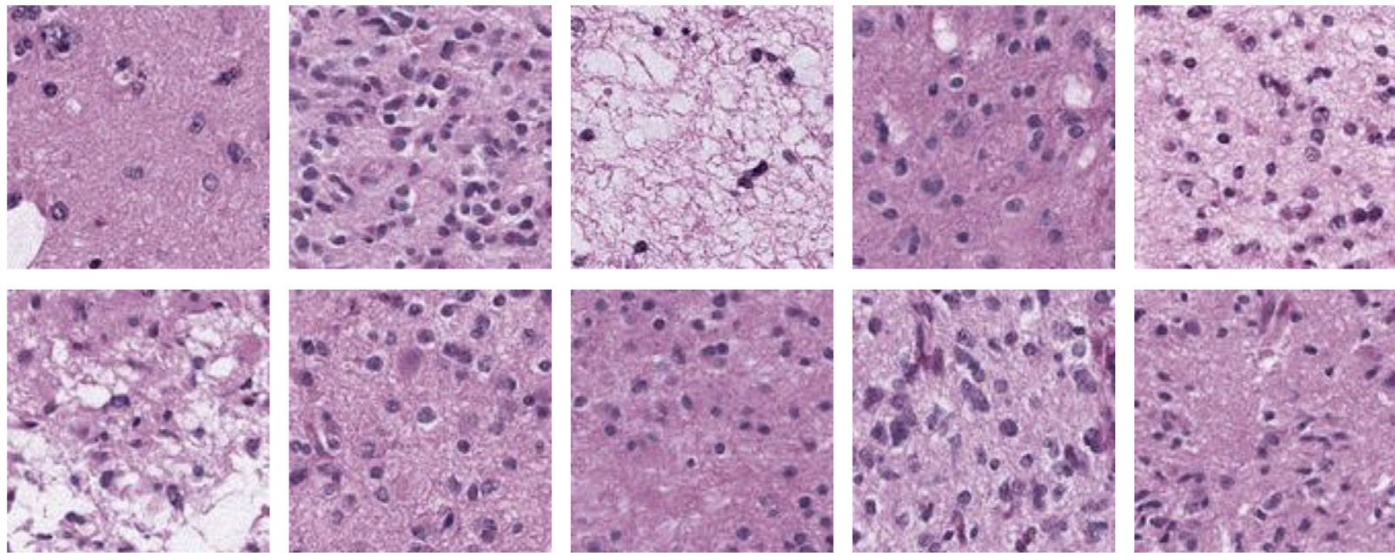
Julia Wolleb



<https://youtu.be/US9CzPrT2H8?si=Rrlt7CYwKE55rd5H>

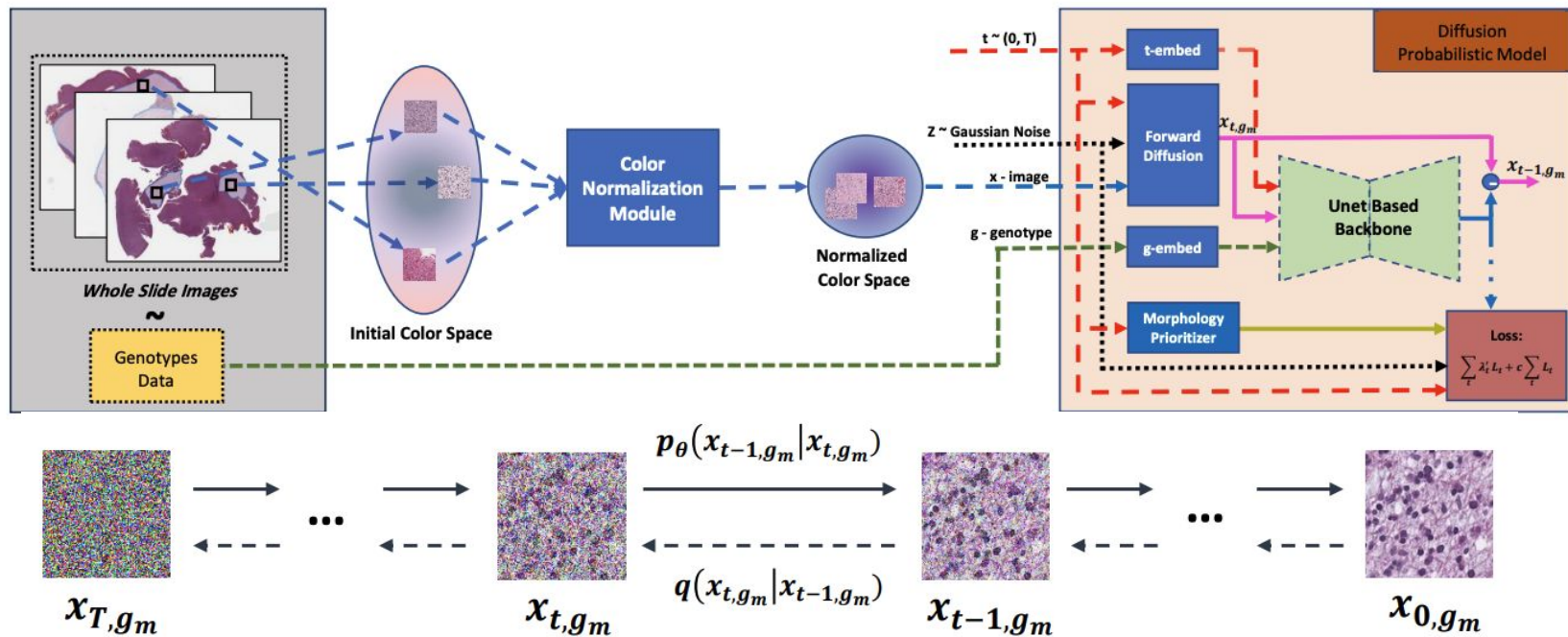
Applications in medical image generation

- Diffusion models have remarkable performance in generating **synthetic medical images**
 - aiding data augmentation and rare disease representation



Applications in medical image generation

- Generating **histopathology** images with **genotype** guidance



Genotype-conditioned Image Generation

How Are IDH Mutation and 1p/19q Co-Deletion Extracted for Training Data?

The training data for the diffusion model includes **paired histopathology images and genotype information**. The genotype labels (IDH mutation status and 1p/19q co-deletion) are extracted through **molecular and genetic tests**, including:

1. IDH Mutation Detection

- **Immunohistochemistry (IHC)**: Uses **antibodies** to detect **mutant IDH1 protein** in tumor samples.
- **Next-Generation Sequencing (NGS)**: Directly sequences the **IDH1/IDH2 genes** to detect mutations.

2. 1p/19q Co-Deletion Detection

- **Fluorescence In Situ Hybridization (FISH)**: A cytogenetic test that detects the **loss of chromosome arms 1p and 19q** in tumor cells.
- **Comparative Genomic Hybridization (CGH)**: Identifies chromosomal deletions.
- **PCR-based methods**: Detects loss of heterozygosity (LOH) in 1p and 19q regions.

Once the genetic data is obtained, it is **paired with corresponding histopathology images** to create a **genotype-labeled dataset** for training.

Genotype-conditioned Image Generation

- **What is an IDH Mutation?**

- IDH (Isocitrate Dehydrogenase) Mutation refers to genetic alterations in the IDH1 or IDH2 genes.
- These mutations are commonly found in gliomas (brain tumors) and are important for tumor classification, prognosis, and treatment decisions.
- IDH-mutant gliomas tend to have better survival rates compared to IDH-wildtype gliomas, which are more aggressive.

- **What is 1p/19q Co-Deletion?**

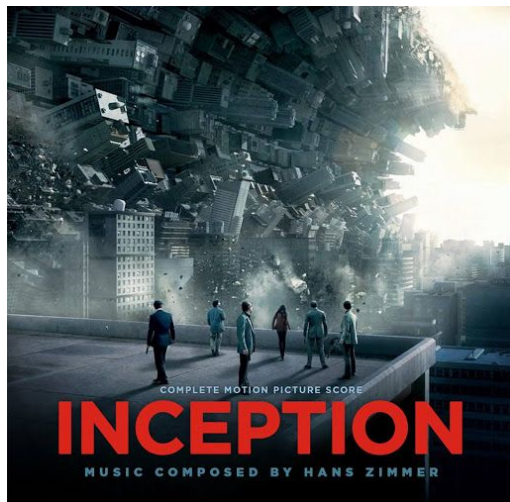
- The 1p/19q co-deletion is a chromosomal alteration where parts of chromosomes 1p (short arm of chromosome 1) and 19q (long arm of chromosome 19) are missing.
- This is a key molecular marker used to classify gliomas.
- Gliomas with 1p/19q co-deletion are almost always oligodendrogliomas, which respond well to chemotherapy and radiation therapy.
- If a glioma has IDH mutation but no 1p/19q co-deletion, it is classified as an astrocytoma instead of an oligodendroglioma.

Inception score

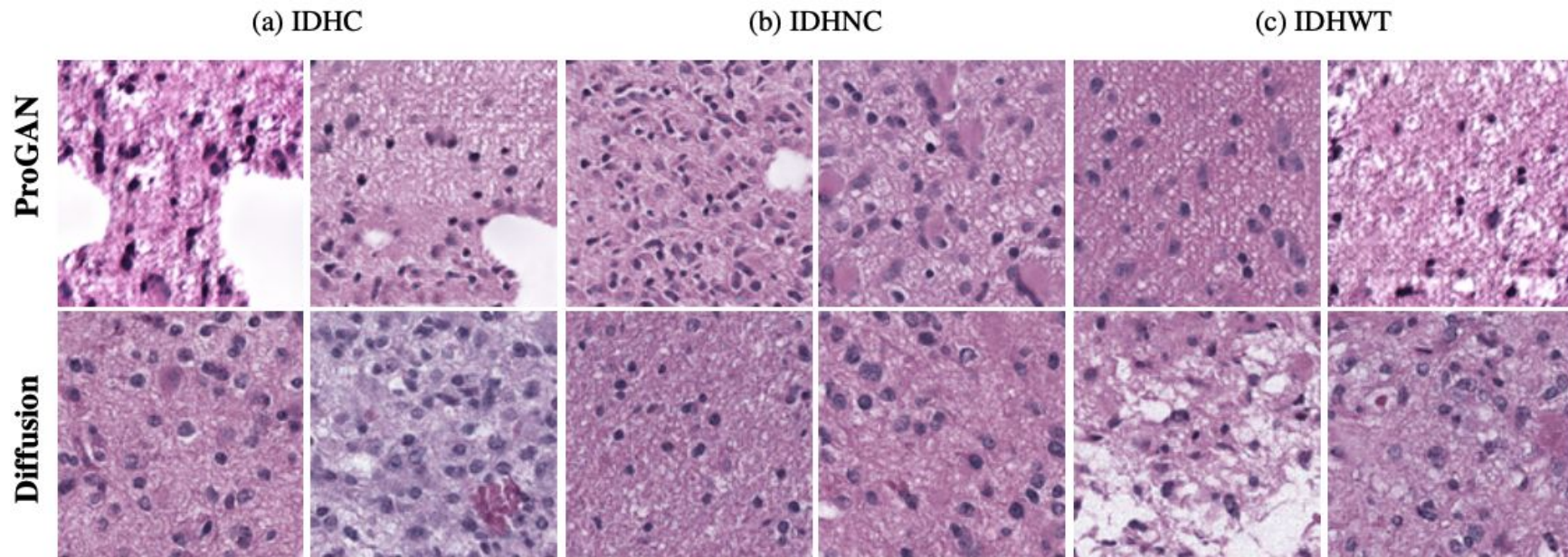
What it measures: The **quality** and **diversity** of generated images.

How it works:

- Uses a pre-trained **Inception network** (e.g., Inception v3) to classify generated images.
- Measures:
 - **Quality:** If a generated image is highly classifiable (i.e., strong class predictions).
 - **Diversity:** If the generated images cover multiple classes.
- Requires a **well-trained classifier on medical data**, unlike natural images (e.g., CIFAR, ImageNet).



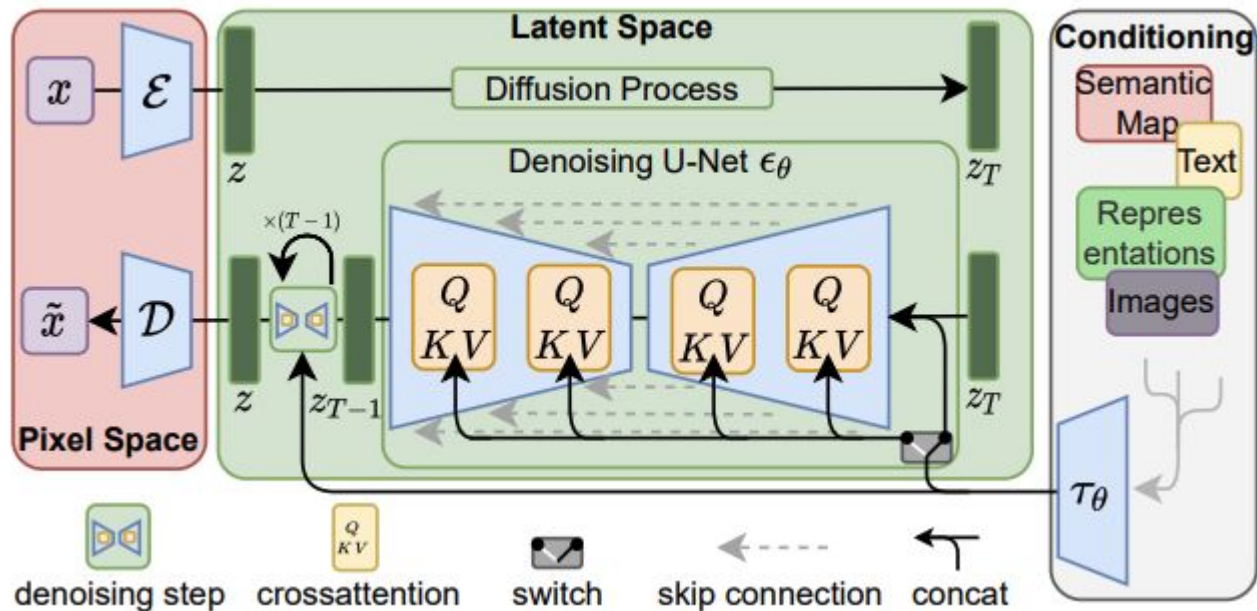
Applications in medical image generation



	ProGAN	Diffusion Model
Improved Recall	0.4816	0.8528
Improved Precision	0.0078	0.2573

	ProGAN	Diffusion Model
Inception Score	1.67	2.08
FID	53.85	20.11
sFID	24.37	6.32

Latent-space Conditional Diffusion



<https://arxiv.org/abs/2112.10752>

Self-Attention Module

- **Purpose:** Captures long-range dependencies within feature maps.

Mechanism:

- Flattens spatial dimensions into “tokens.”
- Applies multi-head self-attention (along with layer normalization and a feedforward network) to let every part of the feature map attend to every other part.
- **Role in Diffusion:** Helps preserve global structure during the denoising process.

Self-attention vs Cross-attention

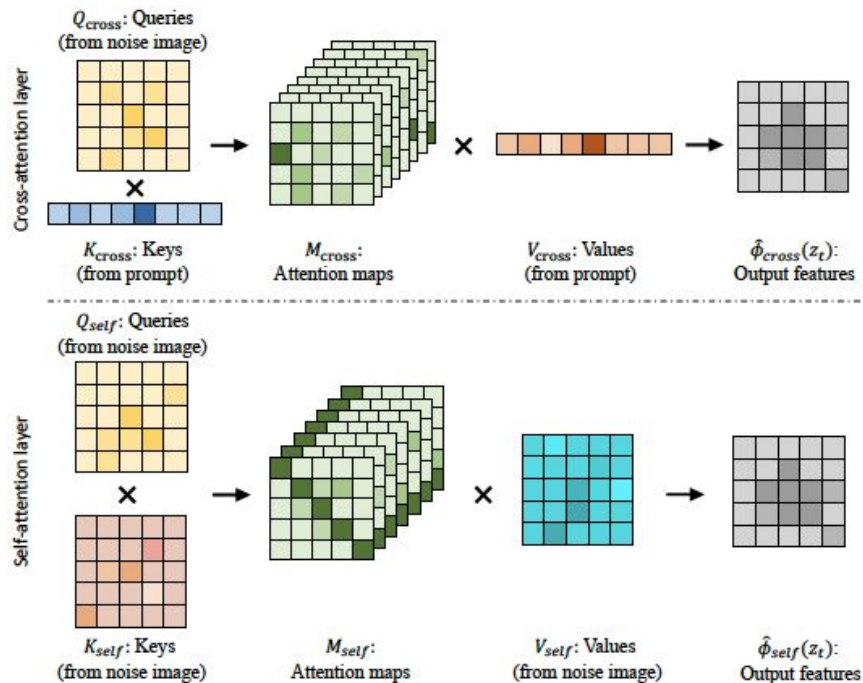
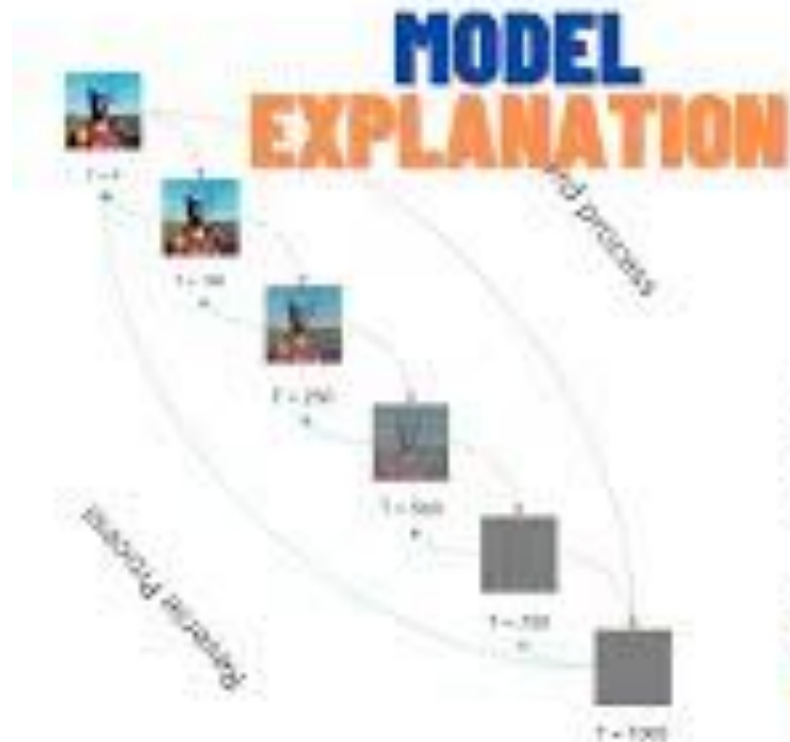


Figure 2. Cross and self-attention layers in Stable Diffusion.



```

def denoise_fn(x, t, c):
    """
    Denoise function.
    """
    # Extract noise level
    noise_level = t

    # Extract class label
    class_label = c

    # Denoise the image
    x = denoise_fn(x, noise_level, class_label)

    # Return the denoised image
    return x
  
```

Denoising Diffusion Probabilistic Models (DDPM)

Umar Jamil

Full TRAINING CODE